

Retail Customer Behavior & Sales Intelligence

End-to-End Data Analytics Portfolio Project

Prepared as an independent portfolio implementation using Python, SQL and Power BI. The workflow is inspired by the referenced tutorial, while the analysis, documentation and deliverables here are independently organized for a Data Analyst portfolio.

Executive Summary

The dataset contains 3,900 records and 3,900 unique customer IDs. Total recorded purchase value is \$233,081, with an average purchase of \$59.76. The overall average review rating after median imputation is 3.75.

Business Problem

The retail business needs a practical view of customer purchasing behavior to support product, promotion, retention and subscription decisions. The analysis therefore combines data quality work, exploratory analysis, SQL business questions and an executive Power BI dashboard.

Data Preparation

The raw data contains 37 missing review ratings and no duplicate rows. Missing ratings were imputed with the overall median rating. Age bands, customer segments, discount flags and promo flags were created for analysis.

Key Findings

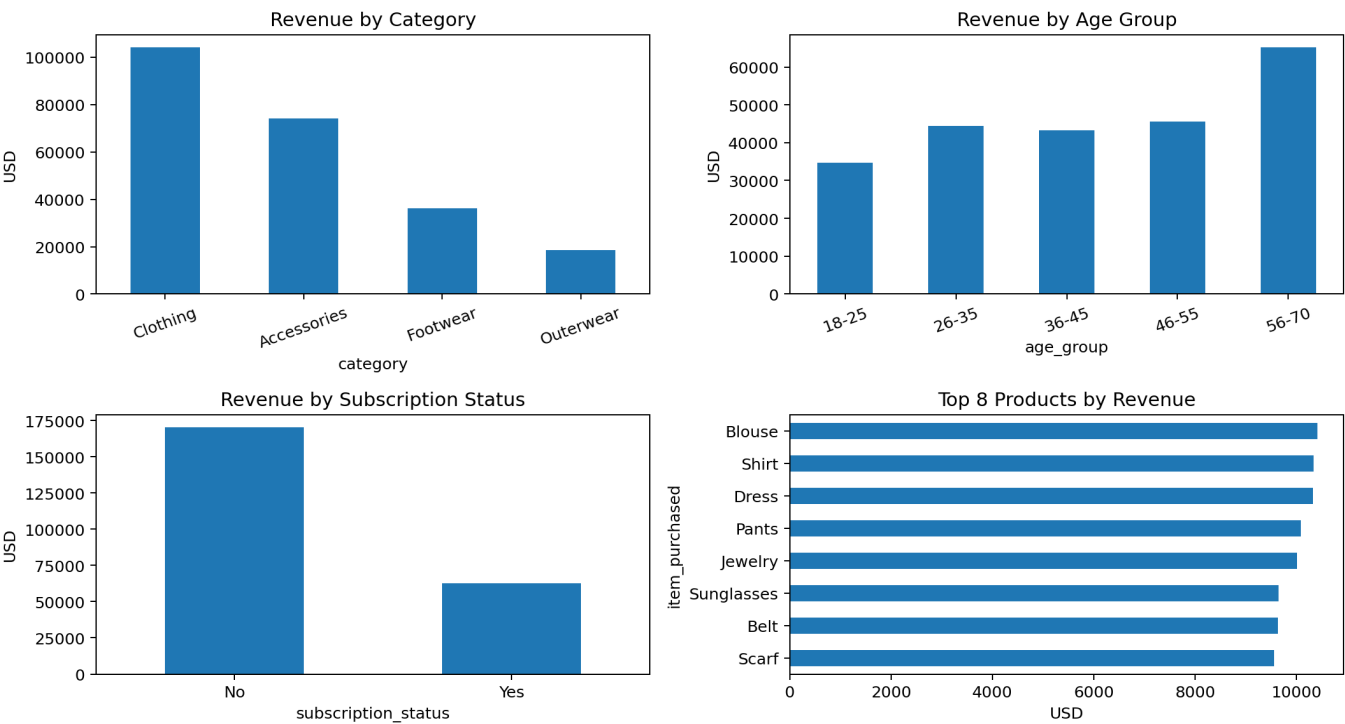
- Clothing: \$104,264 revenue across 1,737 orders.
- Accessories: \$74,200 revenue across 1,240 orders.
- Footwear: \$36,093 revenue across 599 orders.
- Outerwear: \$18,524 revenue across 324 orders.
- The 56–70 age group contributes the largest revenue: \$65,256.
- Loyal customers account for 3,116 records and \$185,517 in recorded revenue under the project segmentation rule.
- Subscribers represent 27.0% of records and generate \$62,645 in recorded revenue.
- 43.0% of records have a discount applied.

Recommendations

1. Prioritize high-revenue categories and products for inventory and merchandising reviews.
2. Monitor subscription conversion and retention separately rather than assuming subscribers spend more.
3. Review products with high discount rates for margin impact and promotion dependency.
4. Use customer segments to tailor retention campaigns and repeat-purchase offers.
5. Add a validated transaction date in future data collection so the business can measure monthly trends and seasonality over time.

Dashboard Preview

Retail Customer Behavior — Dashboard Preview



Technical Stack

Python (Pandas, NumPy, Matplotlib) • PostgreSQL SQL • Power BI • Excel • Git/GitHub

Interview Note

The strongest interview discussion is not the dashboard alone. Explain the business question, data-quality checks, segmentation logic, SQL analysis, dashboard design choices, findings, limitations and what you would do with richer production data.